

Simulation development portfolio



Our development capability

We bring together simulation software, editable 3D content, immersive training and hardware integration. Our experience spans emergency response, command exercises, VR applications and naval simulation development.

Application to RMN

Our proposed RMN delivery combines ship and scenery authoring, instructor tools and maintainable training content with Malaysian development, integration and lifecycle support.

Company-supplied VLNAVSIM presentation, 2025: training equipment photograph. Portfolio sections cover company experience, interface design and current development studies.

Fire safety and command training



Coordinated training

Our presentation describes the Zhushan Training Center programme in Taiwan: tabletop, VR and AR training, spatial workflows and software/hardware integration, with coordinated exercises for up to 20 participants.

Editable scenarios

Command training experience includes incident configuration, unit deployment and orders. These disciplines inform instructor-led scenario development and multi-role training for maritime applications.

VR and physical interaction



Digital and physical interaction

Our portfolio includes vehicle driving with motion-based hardware, drone training with dedicated controllers and weather-recognition exercises. Training props and control interfaces connect the virtual environment with practical interaction.

Integration discipline

The same development disciplines support control mapping, user feedback and coordinated simulation states. RMN-specific interfaces and hardware integration remain subject to the agreed design and validation programme.

Ship model development



Editable vessel assets

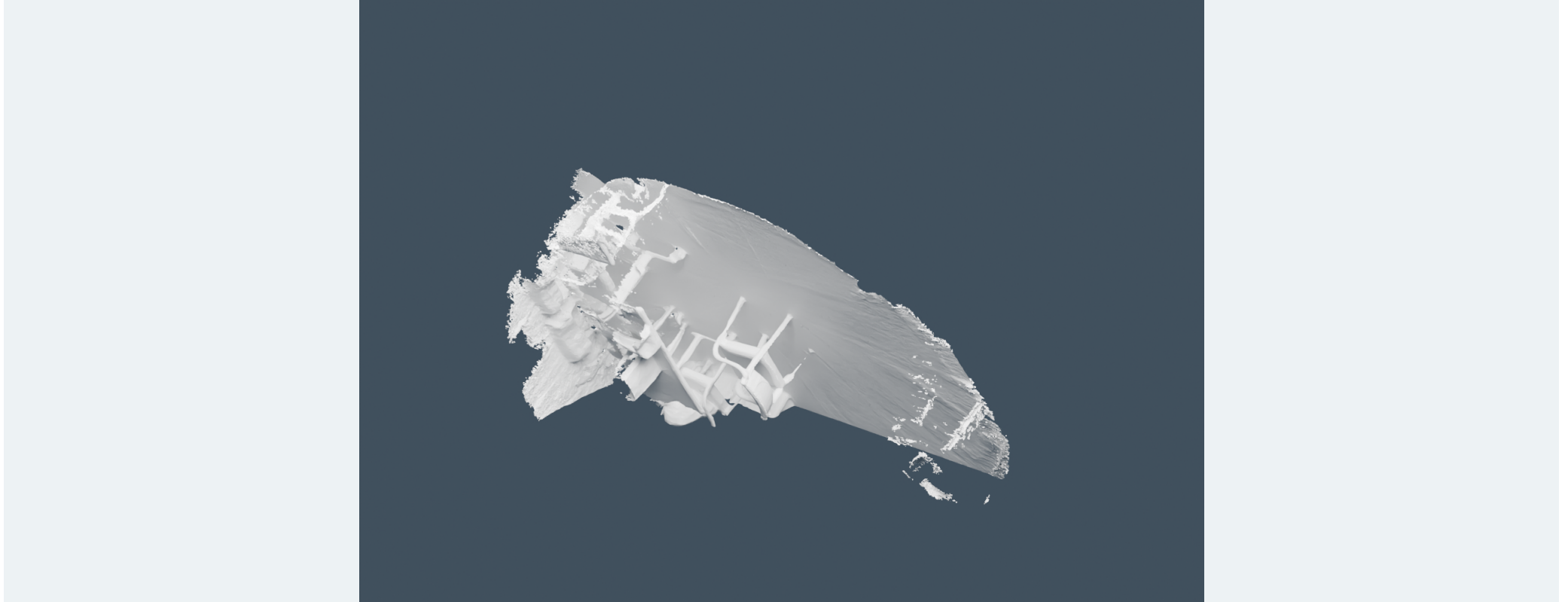
Our current vessel study contains named components, editable geometry and materials, with Blender source and GLB export. It demonstrates a visual authoring workflow that can support subsequent modification and simulator content integration.

Production workflow

For a project vessel, references and dimensional inputs guide modelling, optimisation and review. The present study is a concept visual model; ship-specific dimensions, fidelity and handling parameters require separate validation.

Current in-house development study: DEV_SHIP_001. Editable source and exported asset are available in the detailed project portfolio.

Captured content and optimisation



Processing workflow

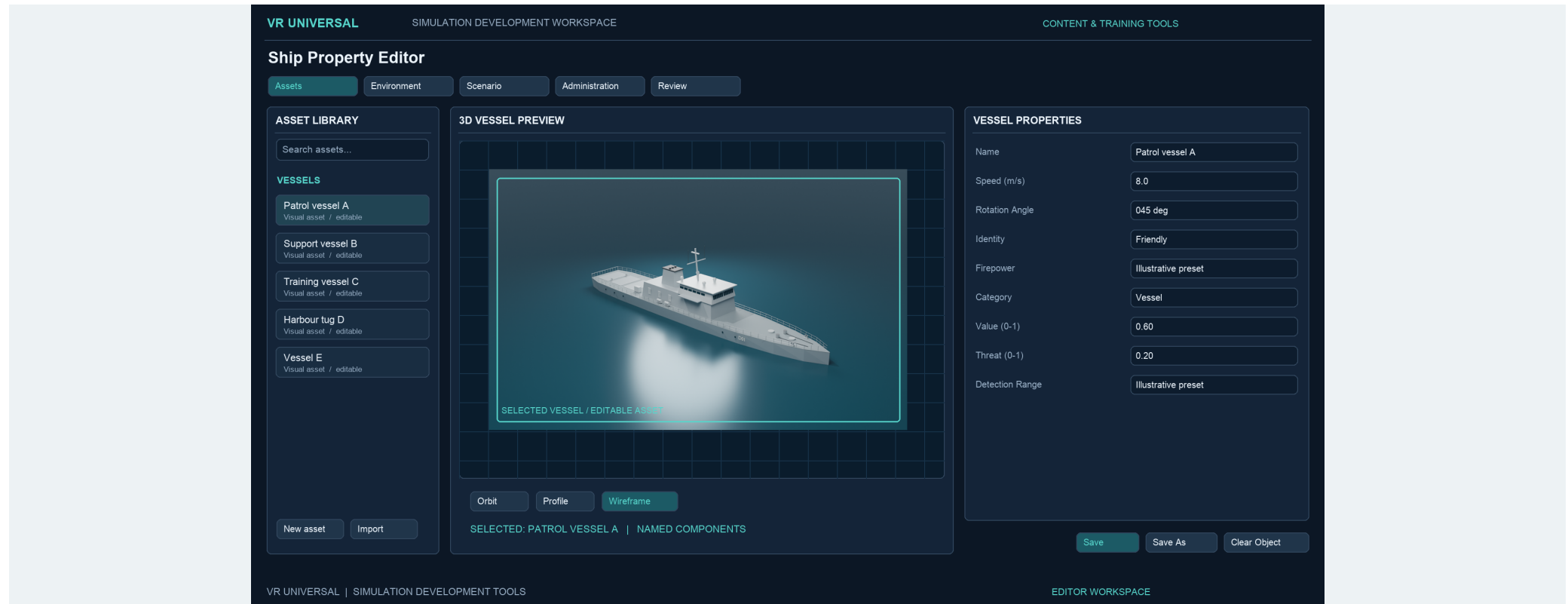
Our public scan study demonstrates mesh inspection, simplification and export. The source contains 935,255 faces; the optimised derivative contains 139,968 faces, a reduction of approximately 85%.

Use in simulator content

This workflow supports preparation of captured objects for interactive environments. The demonstration uses an existing public scan; it does not claim that we captured this dataset or validated its dimensions.

Public scan source: Choi, Zhou, Miller and Koltun, Redwood dataset (2016). Attribution and processing records accompany the study.

Ship Property Editor



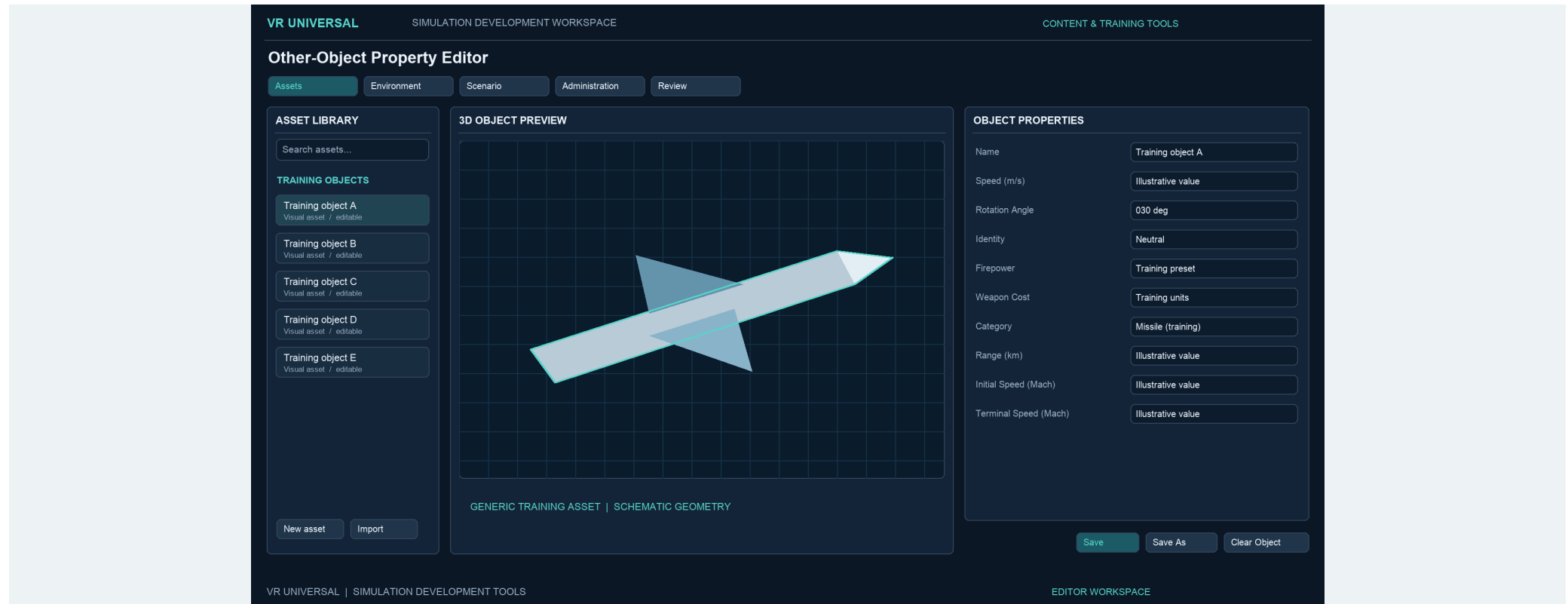
Vessel configuration

Our vessel editor design brings together asset selection, a 3D preview and configurable name, speed, rotation, identity and category. Vessel-specific fields include firepower, value, threat and detection range.

Content authoring workflow

The workspace includes search, Save, Save As and Clear Object controls. A selected-object outline and viewing controls explain how vessel content can be inspected and configured for the training library.

Other-Object Property Editor



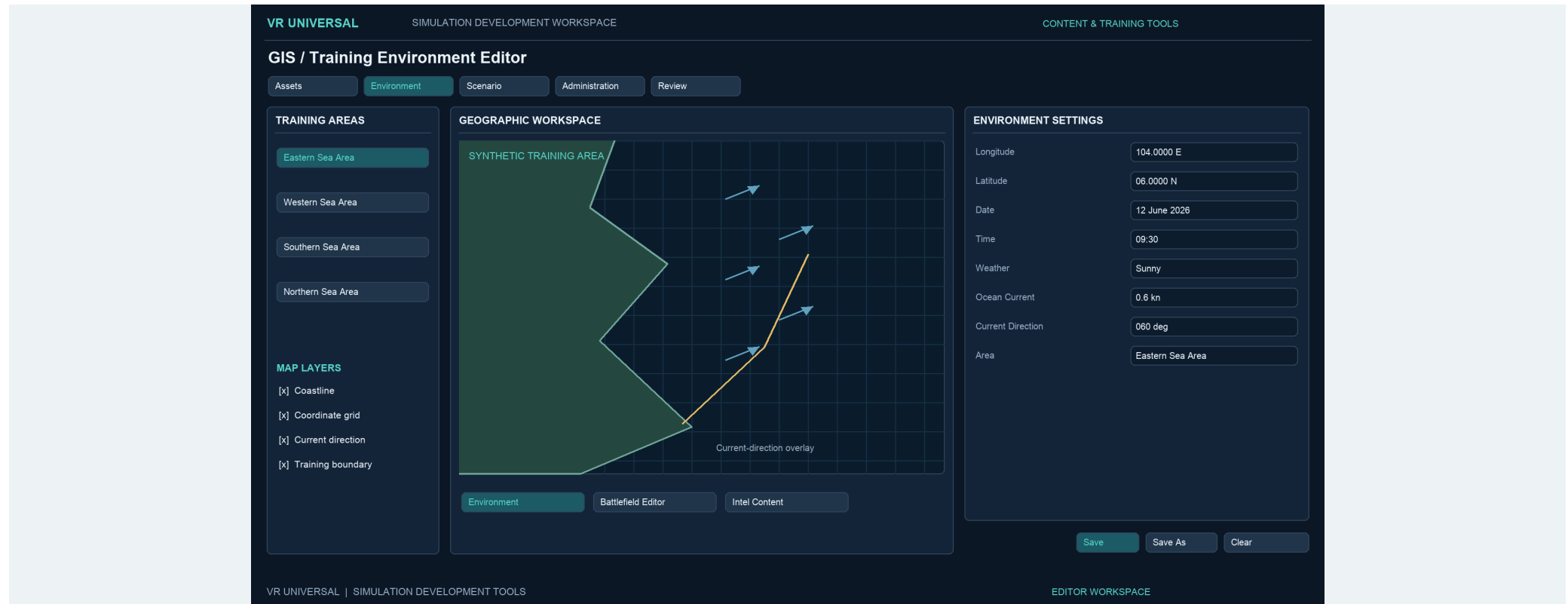
Category-specific configuration

Our object editor design separates the asset library, geometry preview and parameter panel. The training-object example includes identity, rotation, speed, firepower, weapon cost, range and initial/terminal speed fields.

Reusable authoring structure

The workspace presents search, selection and save controls within one editing workflow. Object categories and their parameters are adapted to the agreed training scope; this page illustrates the category-specific structure.

GIS and Training Environment Editor



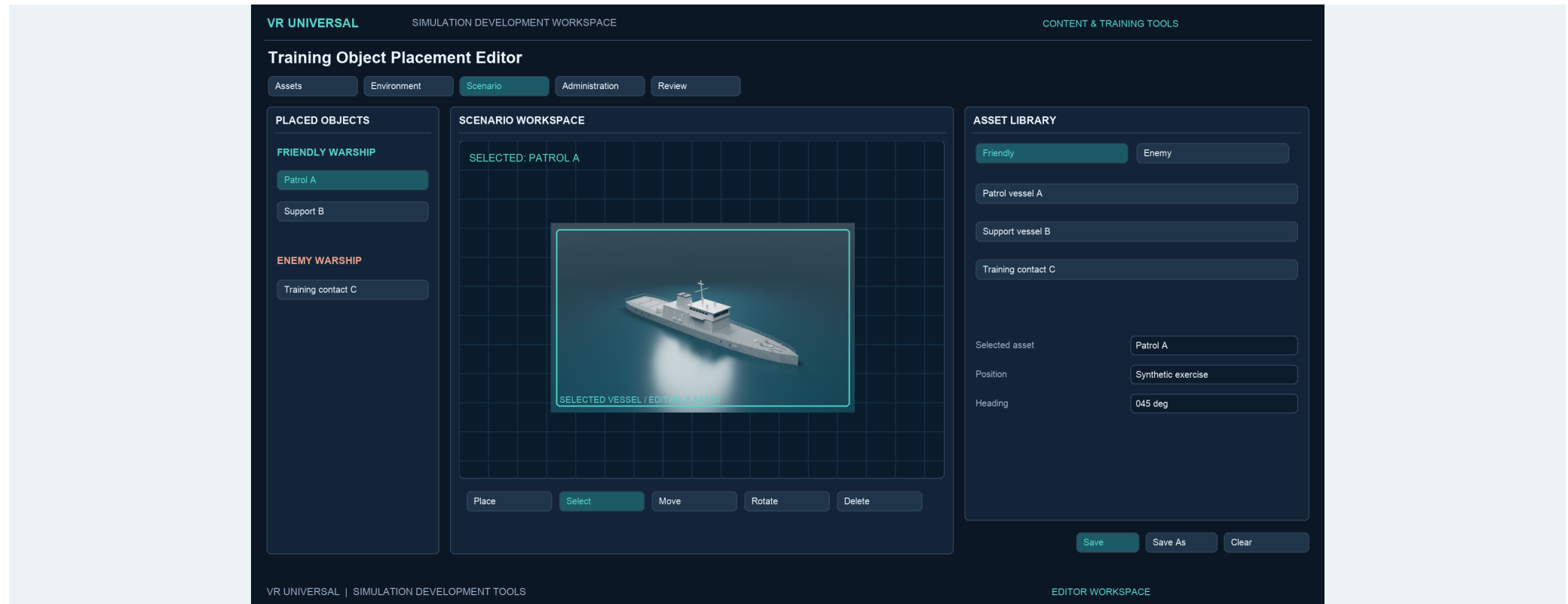
Geographic exercise configuration

Our environment editor design combines regional area selection and a map workspace with longitude, latitude, date, time, weather and ocean-current settings. Layer controls and current arrows clarify the configured training environment.

Connected scenario tools

Environment, Battlefield Editor and Intel Content tabs connect area configuration with object placement and scenario information. Save, Save As and Clear support the authoring workflow. The map is synthetic; production GIS data and geographic accuracy require validation.

Training Object Placement Editor



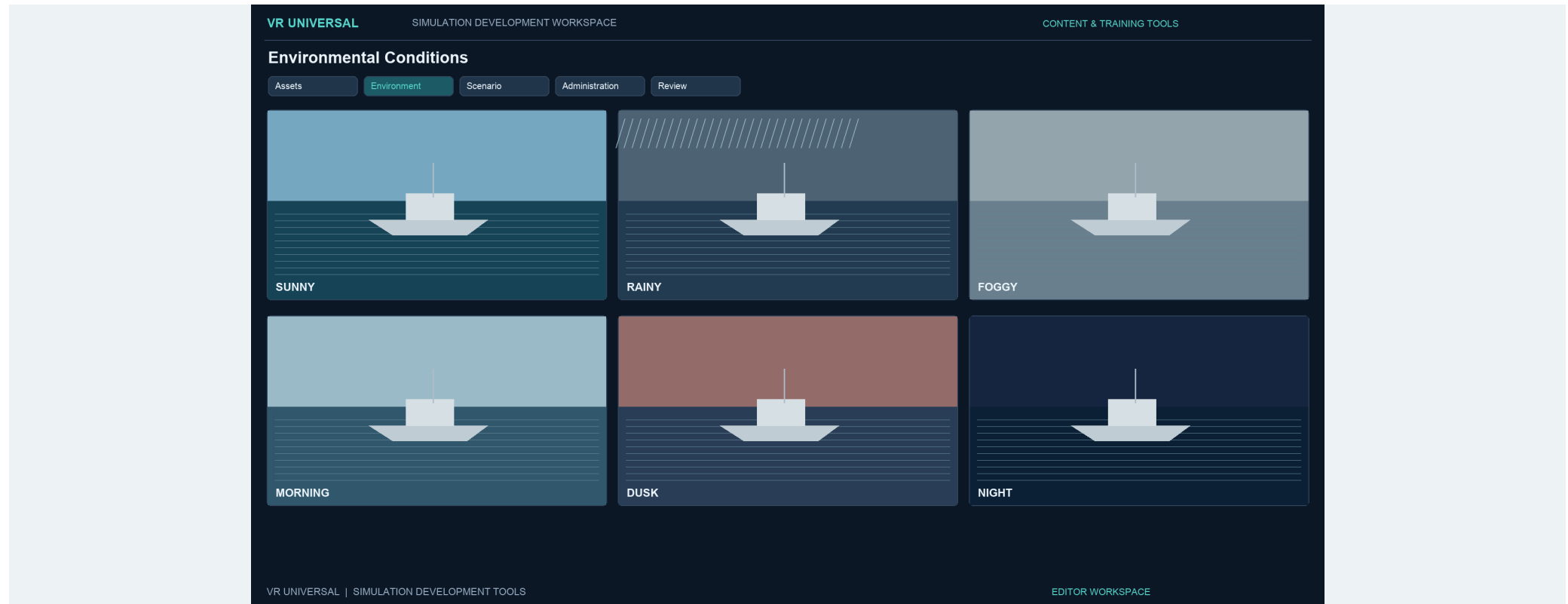
Scenario composition

The placement workspace combines friendly/enemy object groups, an asset library and a selected vessel preview. Placement, selection, deletion and saving controls explain scene assembly; move and rotate controls are included in this interface design.

RMN content independence

Asset definition and scenario placement are presented as separate editing tasks. Our RMN delivery plan provides the agreed source content, developer tools and documentation to support independent scenario modification.

Environmental conditions



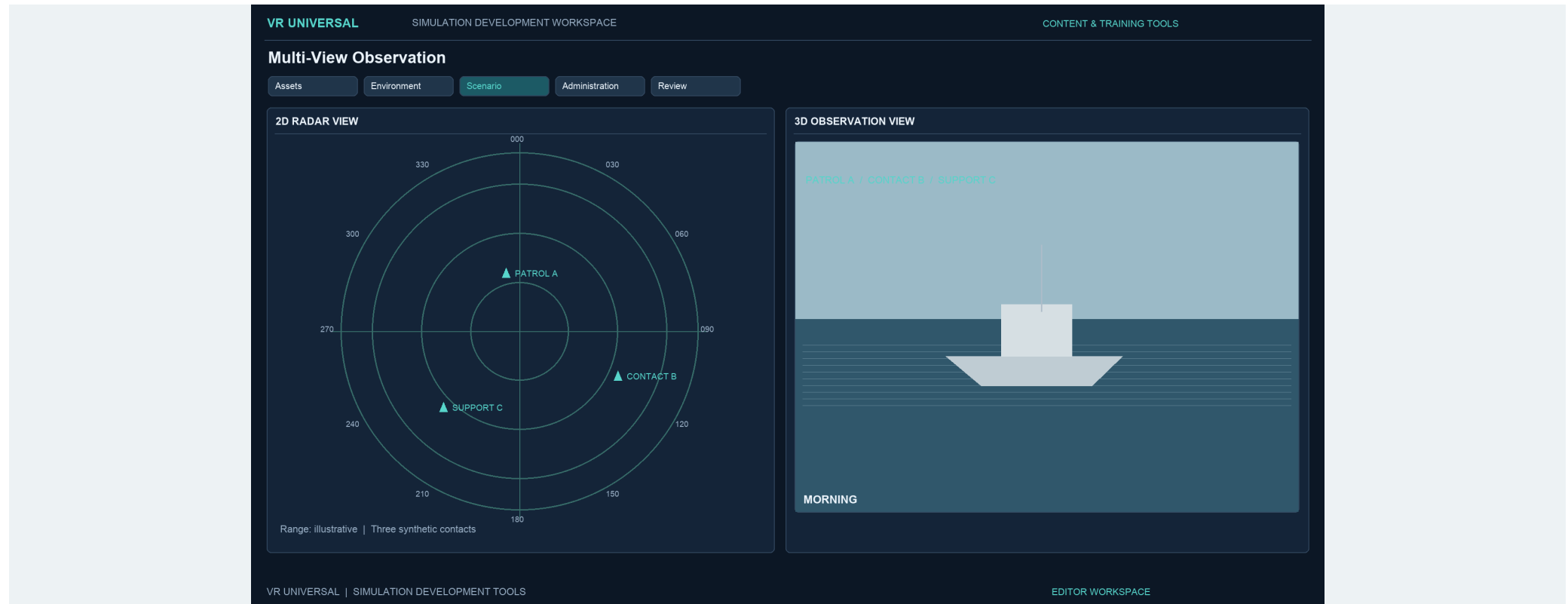
Weather and lighting

The interface presents sunny, rainy and foggy conditions alongside morning, dusk and night. These visual studies illustrate the available presentation choices within an instructor-configured exercise.

Training application

Visibility and lighting variations support situational-awareness training. The illustrations show the selection concept; production visual quality and consistency with environmental parameters are validated through the delivery programme.

Multiple training views



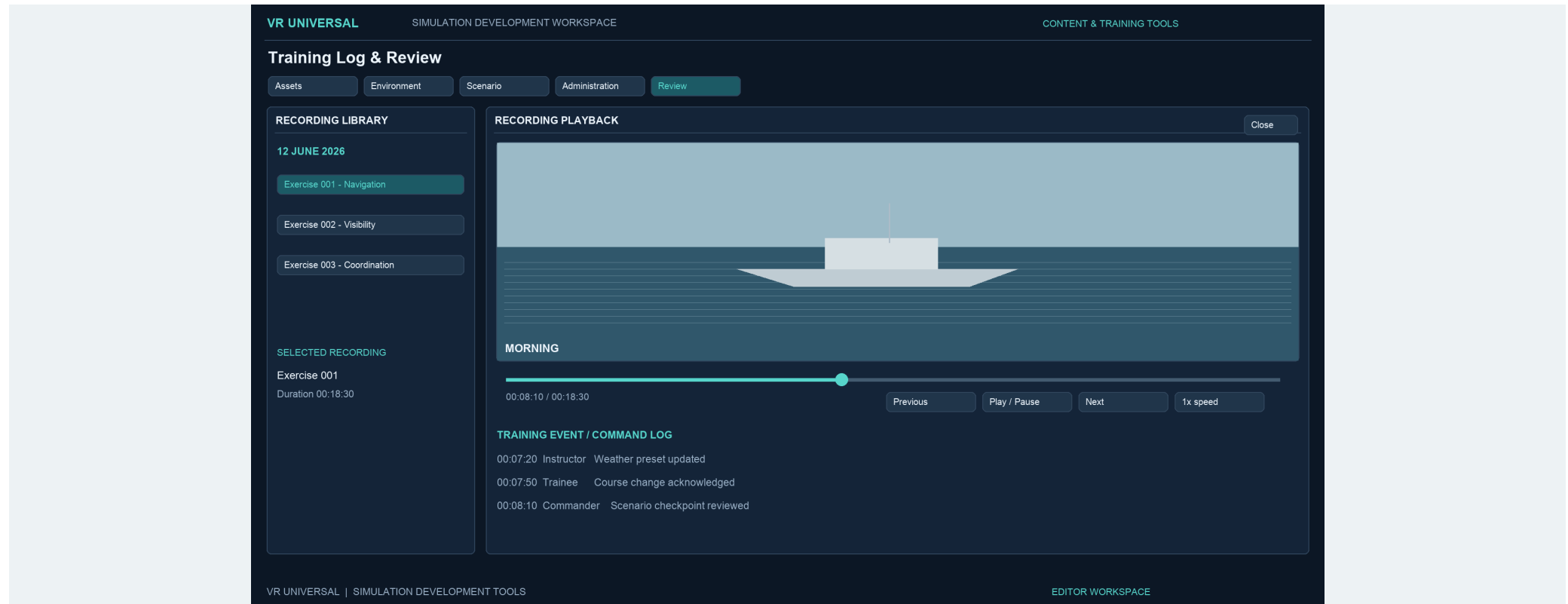
Complementary displays

Our interface design presents a radar-style plan view beside a 3D observation window. The same fictional contacts are identified across the layout to explain how operators can access complementary views of an exercise.

Integration scope

View layouts support coordinated training displays. Equipment-specific radar and navigation behaviour, contact synchronisation and instrument fidelity are defined and validated within the agreed bridge simulator scope.

Training logs and review



Exercise review workflow

The review workspace brings together date-organised recordings, selected playback, a timeline, playback controls and a command/event log. Fictional exercise entries illustrate how instructors can revisit the training sequence.

Debrief and assessment

The interface separates playback from timestamped events to support discussion of decisions and actions. Assessment measures, report outputs, record retention and access controls are agreed for the RMN delivery.

Scenario information and Intel Content

The screenshot displays the 'VR UNIVERSAL SIMULATION DEVELOPMENT WORKSPACE' interface. At the top, there are tabs for 'Assets', 'Environment', 'Scenario' (selected), 'Administration', and 'Review'. The main content area is divided into two columns. The left column is titled 'FRIENDLY PLATFORM' and contains fields for 'Vessel name' (Patrol vessel A), 'Code name' (DEMO-ALPHA), 'Detected longitude' (Synthetic coordinate), 'Detected latitude' (Synthetic coordinate), 'Launcher configuration' (Training preset), and 'Firepower summary' (Illustrative parameters). Below these fields is a section titled 'PLATFORM / CONFIGURATION SUMMARY' with the text 'Scenario information for instructor review'. The right column is titled 'ENEMY / TRAINING CONTACT' and contains similar fields: 'Vessel name' (Training contact C), 'Code name' (DEMO-CHARLIE), 'Detected longitude' (Synthetic coordinate), 'Detected latitude' (Synthetic coordinate), 'Launcher configuration' (Training preset), and 'Firepower summary' (Illustrative parameters). Below these fields is a section titled 'PLATFORM / CONFIGURATION SUMMARY' with the text 'Scenario information for instructor review'. At the bottom of the interface, there are tabs for 'Environment', 'Battlefield Editor', and 'Intel Content' (selected). The footer of the interface shows 'VR UNIVERSAL | SIMULATION DEVELOPMENT TOOLS' on the left and 'EDITOR WORKSPACE' on the right.

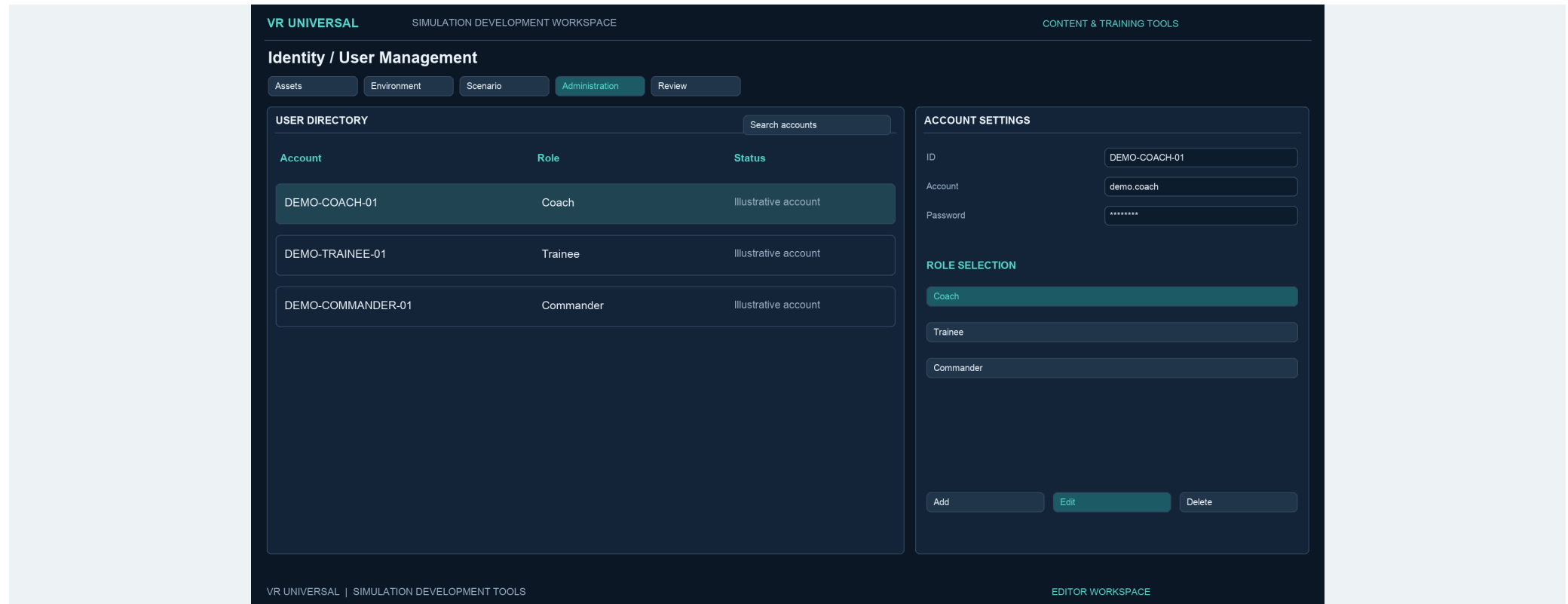
Platform summaries

The scenario information design separates friendly platforms from training contacts. Vessel name, code name, detected coordinates, launcher configuration and firepower summaries provide a structured instructor view.

Connected authoring workflow

Environment, placement and scenario information are presented within the same editing structure. All platform names, coordinates and configurations on this page are fictional examples.

User and role management



Account administration

The design includes a user directory, ID, account and masked-password fields, coach/trainee/commander role selection and Add, Edit and Delete controls. Only fictional demonstration accounts appear.

Delivery requirements

Role-specific access and administration support organised training delivery. Authentication, permission enforcement and secure storage are implementation and validation requirements; the illustration explains the administration workflow.

Local development and RMN independence



Development tools and local delivery

Current demonstrations use Blender, Python, GLB, JavaScript, JSON and SQLite. The VLNAVSIM presentation describes the VL-Simulation Engine approach. RMN-specific design, programming, testing, integration and validation will be undertaken in Malaysia under the SOC requirements.

Handover and capability transfer

Our delivery plan addresses editable ship and scenery content, tool/licence inventories, English documentation, backup/recovery, agreed RMN-content rights and dedicated training for six RMN developers. Third-party components retain their licences; local facility, personnel and access records support the delivery commitment separately.

Current harbour development study. SOC alignment: 2.7.1.6.3.2, 2.7.1.7.5, 3.4.3-3.4.4 and 3.13. Development studies and proposed delivery are distinguished from prior experience.